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Solving Nurse Rostering Optimization Problem using Reinforcement Learning - Simulated Annealing with Reheating Hyper-heuristics Algorithm

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Abstract

Nurse rostering is the process of scheduling hospital nurses for a certain period of time. This is a complex and interesting task that involves multiple perspectives, including those of the hospital, nurses, and patients. In addition, nurse scheduling must consider existing constraints to obtain a schedule that is close to optimal. Common constraints include the number of available nurses, their abilities, labor regulations, and hospital policies. The numerous constraints in nurse scheduling lead to a problem known as the Nurse Rostering Problem (NRP), which is difficult to solve conventionally and is considered an NP-hard problem. Many researchers have attempted to solve NRP computationally, as this approach reduces the time required to create nurse schedules. The less time needed to create nurse schedules, the more time can be allocated to improving medical services, which ultimately impacts the quality of healthcare in the hospital's surrounding area. Several approaches have been used to solve NRP, including hyper-heuristic methods, which are search methods used to solve optimization problems. The aim of this study is to employ a hyper-heuristic approach using Reinforcement Learning - Simulated Annealing with Reheating algorithm to solve the nurse scheduling problem in hospitals in Norway. The results show that the Reinforcement Learning - Simulated Annealing with Reheating algorithm can produce schedules with an 82% improvement in solutions. This algorithm also outperforms the comparison algorithms, with a 3% to 47% improvement over the Simple Random - Hill Climbing algorithm, a 2% to 12% improvement over the Reinforcement Learning - Hill Climbing algorithm, and a 0.7% to 7% improvement over the Reinforcement Learning - Simulated Annealing algorithm.

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Keywords: Nurse Rostering Problem; Hyper-heursitic; Reinforcement Learning; Simulated Annealing with Reheating

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1. Introduction

Nurse rostering is the process of scheduling hospital nurse work in a certain period of time. In hospital, generating nurse roster is challenging and tedious works. It is because the roster must be generated in weekly basis and in the scheduling process many aspect must be taken into account such as the nurses' abilities, fair scheduling, and applicable laws and regulations [12]. Nurse rostering is a complex and interesting task to observe. There are several perspectives that must be considered in nurse scheduling. From the hospital's perspective, scheduling must consider the efficiency of nurse resources to minimize costs. From the nurses' perspective, scheduling must consider the balance between work schedules and time off. From the patient's perspective, scheduling must consider the competency of nurses to reduce waiting times for patients and improve the quality of medical services [9]. In addition, nurse scheduling must consider existing constraints to obtain a work schedule that is close to optimal. Constraints that often arise in nurse scheduling include the number of available nurses, nurse competency, labor regulations, and hospital policies. The many constraints that must be met will result in a problem that is often called the NRP [19].

To date, the NRP is still considered an unsolvable problem or often called an NP-hard problem [20]. Many studies have attempted to solve the Nurse Rostering Problem computationally using various algorithms with metaheuristic [4, 2], or hyper-heuristic [9] approaches. Solving the Nurse Rostering Problem computationally will reduce the time required to schedule hospital nurses' work. The less time required to schedule nurse work, the more time available to improve medical services. This will have an impact on the quality of health in the surrounding area of the hospital [3].

Several studies have attempted to solve the Nurse Rostering Problem using various approaches. One study that addressed the Nurse Rostering Problem is the study carried out in [18] that used a hybrid approach. This study focused on solving the Nurse Rostering Problem in Norwegian hospitals. The study used the Constraint Programming (CP) and Variable Neighborhood Descent (VND) algorithms. The results of the study showed that the hybrid approach can solve large and realistic problems with a relatively short computation time. Another study that addressed the Nurse Rostering Problem is discussed [10] that use Simulated Annealing (SA) algorithm. This study focused on scheduling hospital nurses by considering three roles of nurses. The aim of this study was to schedule nurses' work to meet demand according to each nurse's role. The results of this study showed that the Simulated Annealing (SA) algorithm produced better results than its comparison algorithm, Hybrid Artificial Bee Colony (HABC). On the other hand, a study on the modification of the Simulated Annealing algorithm using the Simulated Annealing with Reheating (SAR) algorithm was carried on in [5]. The study focused on solving course scheduling problems using the Tabu Search with Sampling and Perturbation (TSSP) algorithm to find an initial solution and the SAR algorithm to improve the solution's quality. The results of the study demonstrated that the algorithm produced good solutions and showed scalable performance to course scheduling problems. Later, in [14] Reinforcement Learning (RL) algorithm was investigated to solve scheduling problem. This research focused on solving the Job Shop Scheduling Problem (JSSP) and the Flow Shop Scheduling Problem (FSSP). The study's results indicated that the RL algorithm is a promising approach to solving scheduling problems.

Compared to prior studies in the related scientific literature, the main contribution of this study is the idea of using hyper-heuristic approach instead of meta-heuristic approach by combining Simulated Annealing with Reheating algorithm as move acceptance strategy and reinforcement learning algorithm as low-level heuristic selection strategy. The Simulated Annealing with Reheating algorithm is used to find better solutions. The concept of reheating will work when the algorithm does not find a better solution in several iterations. Other research also suggests that the Reinforcement Learning algorithm can be combined with other methods. That research combined the Artificial Neural Network method with the concept of Reinforcement Learning [15].

2. Literature Review

2.1. Nurse Rostering Problem

Nurses are one of the components that support the business process in a hospital. Nurses have the duty to care for and serve hospital patients. Indirectly, nurses become an important part of the hospital and require their presence to be available 24 hours a day. This condition can be achieved if nurse scheduling is done correctly. Nurse scheduling

becomes increasingly complex because it must meet the needs of both nurses and patients. The complexity of nurse scheduling is a problem that is often referred to as the Nurse Rostering Problem.

The Nurse Rostering Problem is a problem in allocating nurses to several available work time allocations. The allocation must comply with several existing constraints [4]. Some of the constraints that must be complied with include not scheduling nurses on their day off and scheduling the same nurse at the same time. There are two constraints in the Nurse Rostering Problem: hard constraints and soft constraints. Hard constraints are all the constraints that must be met, while soft constraints are the constraints that are not required to be met and can be violated to produce a solution that can be applied [8].

2.2. Norwegian Hospital Dataset

The Norwegian hospital dataset is a dataset that comes from one of the hospitals in Norway. The dataset is divided into 7 hospital units that are collected and managed by a European research organization called SINTEF [16]. This dataset consists of several parts, namely the staff section, SIF, nurse requirements, constraints, and desired SIF patterns. In this dataset, there are 7 hard which consist of:

- One nurse is allocated to a maximum of one shift per day.
- Shift requirements must be met precisely every day.
- Total working hours for each nurse should not deviate too far from the employment contract.
- Nurses can only work in shifts that match their competencies.
- The break time between two consecutive shifts must exceed the minimum limit set.
- There must be a minimum period of time off every week.
- Do not exceed the maximum weekly working hours limit.

Furthermore, this dataset also has 9 soft constraints consisting of:

- There should not be too many consecutive working days with the same shift category.
- There should not be too many consecutive working days.
- There should not be too few consecutive working days with the same shift category.
- There should not be too few consecutive working days.
- The number of shifts for each category should not deviate too far from the minimum and maximum limits.
- The scheduling should not deviate too far from the nurse's work contract.
- Grouping the nurses' days off.
- Meeting the desired shift pattern.
- There should not be too many undesired shift patterns.

The optimization of nurse rostering in the Norwegian Hospital aims to minimize the penalty value of the generated schedule. This objective can be assessed by using the following calculation:

$$Minf = \sum_{m=1}^{9} K_m p_m \tag{1}$$

Where K_m is the penalty weight and P_m is the penalty value. The penalty value represents the violation value of the soft constraint. There are 9 soft constraint calculations used to calculate the penalty value and 7 hard constraint calculations that can be seen in [17].

2.3. Simulated Annealing

Simulated Annealing with Reheating (SAR) is a development of the Simulated Annealing algorithm. Every better solution will be accepted, but a worse solution will be accepted with a certain probability. This probability is influenced by the predetermined initial temperature. At each iteration, the probability of accepting a worse solution will decrease [7]. The process of reheating the temperature is carried out when there is no better solution for several iterations. This reheating process is done so that the algorithm can explore when there are no better solutions being produced [6].

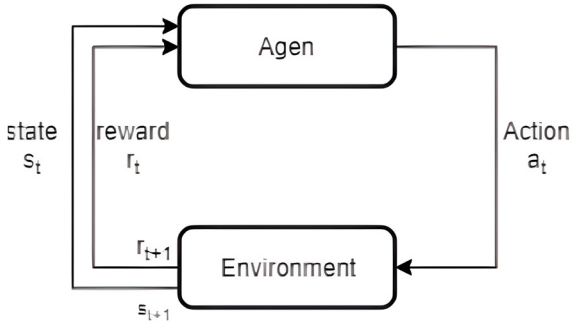


Fig. 1. The Concept of Reinforcement Learning.

2.4. Reinforcement Learning

Reinforcement Learning is a form of machine learning technique where there is an agent (heuristic) that forms an optimal strategy based on the experience of interacting with the external environment. The strategy determines which action to choose for a given state and time. When the agent interacts with an environment and takes an action, the agent enters a new state and receives a reward from that environment [11]. The received reward will be used by the agent to improve its behavior [15]. The working concept of the reinforcement learning algorithm can be seen in Figure 1.

2.5. Hyper-Heuristic

Hyper-heuristics is a search method or learning mechanism for selecting or generating heuristics that can solve combinatorial optimization problems [13]. The selected or generated heuristics are called low-level heuristics, which are categorized into two groups: low-level constructive heuristics and low-level perturbative heuristics. Hyper-heuristics consist of two components: heuristic selection for selecting low-level heuristics, and move acceptance for determining whether changes made to the low-level heuristics are acceptable or not [1].

3. Methodology

3.1. Design Algorithm

In this stage, the design of the algorithm to be used in this research is carried out. This research uses Reinforcement Learning algorithm to select the low-level heuristic and Simulated Annealing with Reheating algorithm for move acceptance, which is shortened to RL-SAR. Both algorithms are combined to find better solutions for the problem domain of nurse scheduling in Norwegian hospitals. The algorithm was chosen because it has been proven effective in some prior studies such as in [7] and [15].

The RL-SAR algorithm is used to optimize the initial solution that has been formed. First, several parameter values are set, such as the initial temperature, cooling rate β, reward, and stopping condition. Then, in the first iteration, the RL-SAR algorithm will randomly select a low-level heuristic. If the low-level heuristic produces a better penalty value, then it will receive a reward, and the penalty value will be saved as Scurr. If Scurr is better than Sbest, then Scurr will become Sbest. Meanwhile, if the low-level heuristic produces a worse penalty value, it will receive a punishment or a reduction in reward. The penalty value generated is not directly rejected. If the probability of accepting a worse penalty value is greater than a random number between 0-1, then the penalty value will be accepted and become Scurr. From the second iteration until the end of the iteration, the selection of low-level heuristics is based on the low-level heuristic with the highest reward value. This process will continue during the specified iterations. With each additional iteration, the initial temperature will continue to decrease according to the cooling rate value.

3.2. Generate Feasible Solution

The initial stage in the nurse scheduling optimization process is to create an initial solution. The initial solution refers to a work schedule that does not violate the existing hard constraints (feasible). The creation of the initial solution begins by reading and storing input data in an array. After all data is read and stored, the next step is to create a matrix that is used to arrange the nurse's work schedule. The matrix will contain the nurse's work schedule according to the length of the scheduling that has been determined. Once the matrix is formed, the next step is to arrange the nurse's work schedule for each day. The scheduling must take into account the existing hard constraints. After the work schedule is formed and feasible, the next step is to calculate the penalty value caused by violations of the soft constraints. The penalty value will be used as data to analyze the performance of the algorithm.

3.3. Optimization Phase

The purpose of solution optimization is to obtain the best penalty value, the smaller the penalty value, the better the solution. Solution optimization is done by exchanging schedules between nurses. This schedule exchange uses several exchange methods, often referred to as low-level heuristics (LLH). There are 3 LLHs used, namely 2-exchange, 3-exchange, and double 2-exchange. Solution optimization is carried out for several specific iterations using the Reinforcement Learning - Simulated Annealing with Reheating algorithm as the main algorithm and several other comparison algorithms.

3.4. Result Analysis

After optimizing the solution, the next step is to analyze the optimization results. The analysis of the optimization results is divided into two stages, namely analysis of the parameters for the main algorithm and analysis of the performance of the main algorithm. In the parameter analysis stage, experiments will be conducted using several parameters to determine the best parameter to be used in the main algorithm. After obtaining the best parameter, the next step is to analyze the performance of the algorithm by comparing it with other benchmark algorithms.

4. Result and Analysis

4.1. Test Environment

In this research, automation and optimization testing were conducted using Intelij and Java programming language. The hardware used was an AMD FX-7500 Radeon R7 laptop with 8 GB of RAM and 240 GB of SSD capacity. The software used was Intellij IDEA and Microsoft Excel as a data storage medium operating on the Windows 10 operating system.

4.2. Initial Solution Result

The initial solution is constructed iteratively by assigning the nurse on for Saturday and Sunday first and then assigning the other available nurses on Monday to Friday. The formation of the initial solution must consider the existing hard constraints. After the solution is deemed to have met all hard constraints, it is accepted and becomes the initial solution. A summary of the preparation of the initial solution for all hospital units can be seen in the following Table 1.

Of the 7 existing hospital units, automatic scheduling was successfully carried out on 3 hospital units, namely Optur 4, Optur 5, and Optur 7. The success of automatic scheduling refers to the resulting work schedule meeting all existing hard constraints. These three work schedules will then be optimized to obtain the smallest penalty value.

There are 4 hospital units that were not successfully scheduled automatically because they still violate several hard constraints. The majority of the violated hard constraints are hard constraint 3 and hard constraint 6. Hard constraint 3 aims to ensure that the working hours of nurses in the resulting schedule do not deviate too far from the contracted working hours. In the implementation process, the four work schedules for hospital units that were not successfully

Table 1. Initial Solution Result

Hospital Unit	Status	Description
Optur 1	Infeasible	Violating hard constraints 3 and 6
Optur 2	Infeasible	Violating hard constraint 3
Optur 3	Infeasible	Violating hard constraint 3
Optur 4	Feasible	Meeting all hard constraints
Optur 5	Feasible	Meeting all hard constraints
Optur 6	Infeasible	Violating hard constraints 3 and 6
Optur 7	Feasible	Meeting all hard constraints

scheduled automatically still have a deviation that is too far from the contracted working hours of nurses. This can be caused by the large number of working hours that must be scheduled. Meanwhile, hard constraint 6 aims to ensure that each nurse has a minimum weekly rest time. In the implementation process, 2 out of 4 hospital units that were not successfully scheduled automatically still have nurses with less rest time than the specified limit.

4.3. Experiment Parameter

To generate an optimally optimized solution, several experiments were conducted in this stage to find better settings for four types of parameters. The first parameter is heuristic selection, where two methods were tested, namely Greedy and Epsilon Decay Greedy. The second parameter is the method for updating the utility by testing sum, average, and discount factor methods. The third parameter is the cooling rate, which was tested with three values of 0.5, 0.85, and 0.99995. The last parameter is the reheating frequency, which was tested with three values of 10,000, 50,000, and 100,000. The result shows that the parameter setting using Epsilon Decay Greedy, sum method, cooling rate with a value of 0.99995, and reheating frequency of 50,000 is the best setting in this case study.

4.4. Optimization Result

Out of the 7 datasets available, only three datasets were found to have feasible solutions, namely Optur 4, 5, and 7. These three datasets were optimized by running the RL-SAR algorithm. The optimization results are shown in Figure 2. From these results, it can be seen that the RL-SAR algorithm produced a significant improvement in the solutions. This is evident from the improvement in the Optur 4 and Optur 5 datasets. The Optur 4 dataset saw an improvement in the solution of over 76%, while the Optur 5 dataset saw an improvement of over 81%. Meanwhile, the Optur 7 dataset only saw an improvement of less than 8%.

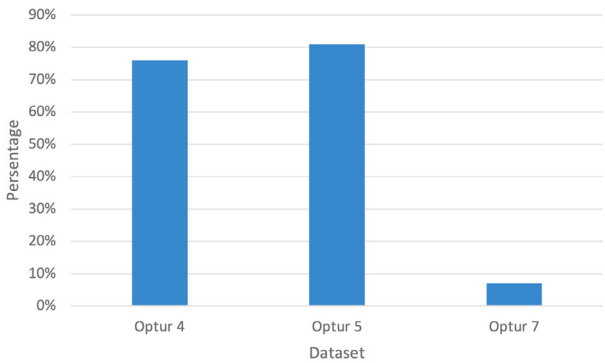


Fig. 2. Optimization Result

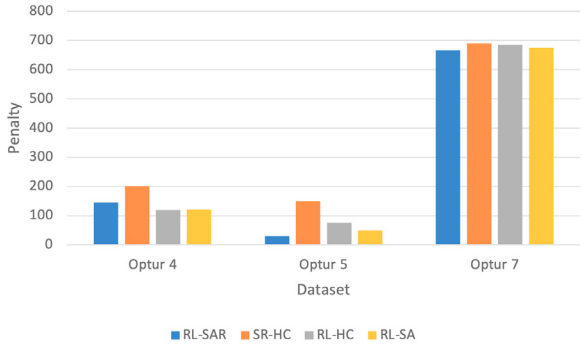


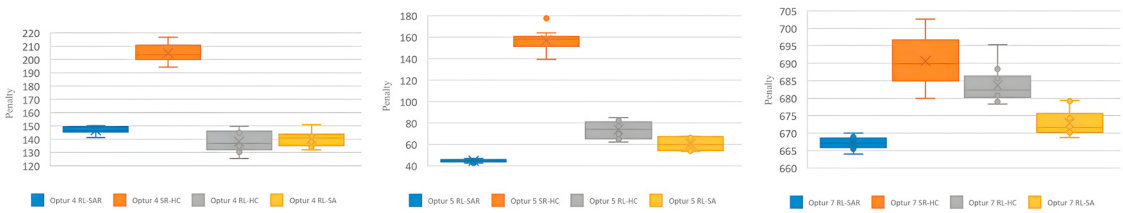
Fig. 3. Comparison Result

4.5. Algorithm Analysis

The performance of the RL-SAR algorithm can be seen by comparing the penalty value of the initial solution with the penalty value of the RL-SAR algorithm. This is done to determine how much improvement is achieved by the RL-SAR algorithm. In this study, the RL-SAR algorithm will be compared with the Simple Random - Hill Climbing (SR-HC) algorithm, the Reinforcement Learning - Hill Climbing (RL-HC) algorithm, and the Reinforcement Learning - Simulated Annealing (RL-SA) algorithm. Each algorithm will be tested 10 times for each hospital unit. The comparison between the RL-SAR algorithm and other algorithms can be seen in Figure 3.

In Figure 3, it can be seen that the RL-SAR algorithm can compete with its comparison algorithms. This is shown by the penalty values produced. The RL-SAR algorithm is able to produce better penalty values than its comparison algorithms for Optur 5 and Optur 7 hospitals. Meanwhile, for Optur 4 hospital, the RL-SAR algorithm can produce better penalty values than the SR-HC algorithm, but it has not been able to produce better penalty values than the RL-HC and RL-SA algorithms. However, the difference in penalty values produced by the RL-SAR, RL-HC, and RL-SA algorithms is not far apart.

Another analysis that can be done from these results is the analysis of the distribution of penalty values. Figure 4. shows the difference in penalty value distribution among the four algorithms. From these results, it can be seen that the penalty value distribution generated by the RL-SAR algorithm shows better results with solutions having a relatively tight distribution of values. On the other hand, the SR-HC, RL-HC, and RL-SA algorithms have penalty value distributions that are diverse and tend to be inconsistent. This can be seen from the box plot diagram formed, where the wider the box plot, the more diverse the penalty values.



(a) (b) (c)

Fig. 4. Comparison of Penalty Value Distribution

5. Conclusion

In conclusion, the study found that the initial solution algorithm used in the nurse rostering problem was not generic enough to create feasible schedules for all hospital units. However, the RL-SAR algorithm was able to successfully solve three dataset from the nurse rostering problem in Norwegian hospitals. The algorithm produced solutions with an 7% to 81% improvement over the initial solution. Additionally, the RL-SAR algorithm outperformed its comparison algorithms (SR-HC, RL-HC, and RL-SA) with a 3% to 47% improvement over SR-HC, 2% to 12% improvement over RL-HC, and 0.7% to 7% improvement over RL-SA. The algorithm's performance was also evident in its ability to produce more consistent results compared to other algorithms. The future works direction could focus on exploring new algorithm strategy as well as testing the algorithm over another nurse rostering problem instances with different characteristics and constraints. Finally, the ability to produce better nurse timetable in hospitals could make the hospital operations more efficient and improve the nurses satisfaction which in turn could improve the hospitals service quality.

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